

Can WANG

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Education

The Hong Kong Polytechnic University	MSc in Urban Informatics and Smart Cities	2022.9 - 2024.10
Capital Normal University	BSc in Geographic Information Science	2018.9 - 2022.6

AWARDS: OUTSTANDING GRADUATE (2022), UNIVERSITY FIRST-CLASS SCHOLARSHIP (2021), UNIVERSITY THIRD-CLASS SCHOLARSHIP (2020)

Research Interests

Multimodal Learning, Visual Grounding and Reasoning, AI-Generated Image/Video Forensics, Image/Video Generation

Academic Papers

LaP-Forensics: Latent-Pixel Consistency Guided Multimodal Reasoning for Deepfake Detection [arXiv:2607.25962]	FIRST AUTHOR, CCF-A (ACM MM 2026)
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#Deepfake Detection/Localization, #MLLM, #DDIM, #SAM, #SFT, #GRPO 2026

- **Problem & Idea:** Addressed reasoning hallucination in MLLM-based AI-generated image forensics, where visually realistic synthetic images can trigger plausible but false artifact explanations; proposed grounding forensic reasoning in latent-pixel consistency that beyond surface RGB cues.
- **Method:** Designed a dual-stream framework that fuses RGB semantic features with DDIM inversion-reconstruction residuals; structured Where-What-Why CoT reasoning produces [SEG] representations that are converted into prompts for SAM-based artifact localization.
- **Result:** Achieved 72.19% mIoU / 63.62% F1 on SynthScars, with +9.14 mIoU over SFT-only and +10.36 mIoU over the w/o forensic-stream variant; the standalone detector improved average UniversalFakeDetect accuracy by +8.24 points over baseline method.

Project/Research Experience

AI Dance Video Pose Transfer Generation Based on Wan2.1	INDUSTRY RESEARCH PROJECT
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#Wan2.1, #LoRA, #DiffSynth, #Diffusion Video 2025.8 - 2025.10

- Built a 5,000+ video dance dataset and fine-tuned Wan2.1-VACE with DiffSynth/LoRA using DeepSpeed on an 8×RTX 4090 cluster, achieving improved pose fidelity and identity consistency

Real-Time 3D Hand Pose Estimation via Asynchronous RGB-Event Fusion	RESEARCH PROJECT, DVSENSE CO.
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#3D Reconstruction, #Pose Estimation, #Event Camera, #Multimodal Fusion, #MANO 2024.9 - 2025.3

- Designed an asynchronous RGB-event fusion framework with a cached slow branch for multi-scale frame-guided hand structure features and a fast event-buffer branch for high-frequency motion cues between RGB frames.
- Combined deformable spatiotemporal alignment with MANO-based two-hand parameter prediction to produce temporally responsive 3D joints and meshes, improving MPJPE by 5 mm / 52% over a single-RGB baseline while reducing inference latency by 20 ms.

Building Change Detection on Panoramic Street View Images	MASTER'S GRADUATION DISSERTATION
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#SVCD, #YOLO, #ViT, #Feature Matching, #End-to-End Detection 2023.8 - 2024.4

- Formulated street-view building change detection as instance-level visual inconsistency localization under viewpoint shift and occlusion; built a 3k+ image-pair / 10k+ instance benchmark with Siamese ViT-feature matching, improving recall by 37% and accuracy by 16%.

Appointments

Algorithm Engineer	DVSENSE (BEIJING) TECHNOLOGY CO., LTD.	2024.9 - Present
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- Developed multimodal RGB-event reconstruction pipelines and a Qwen3-based RAG customer-service system, covering dataset and knowledge-base construction, model validation, robustness testing, and response-quality evaluation.

LLM Evaluation Intern	SENSETIME	2024.5 - 2024.8
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- Evaluated SenseNova outputs with a five-axis rubric; analyzed failure modes and expanded tests for multi-constraint instructions, long-form factual consistency, and multi-turn coreference; integrated the model API into OpenCompass for C-Eval.

Research Assistant	OTTO POON FUND INSTITUTE FOR SMART CITIES, THE HONG KONG POLYTECHNIC UNIVERSITY	2022.8 - 2023.8
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- Developed a remote-sensing land-use conversion and data-annotation system for the GBA Big Data Platform (#DL, #Qt).

Skills

Models & Methods	MLLMs, Diffusion Models, Transformers, CNNs, CLIP, SAM, LoRA, SFT, GRPO, RAG
ML Platforms	PyTorch, HuggingFace Transformers
Programming & Systems	Python, C++, LaTeX, Linux, Git, Docker, ComfyUI
Languages	Mandarin, English (IELTS 7; CET4/6: 550+), Basic Cantonese and French
Multimedia	Photography, Video Editing, Graphic Design, AI-Aided Automation Pipeline for Media Production
Hobbies	Basketball, Snowboarding, Cycling, Kayaking